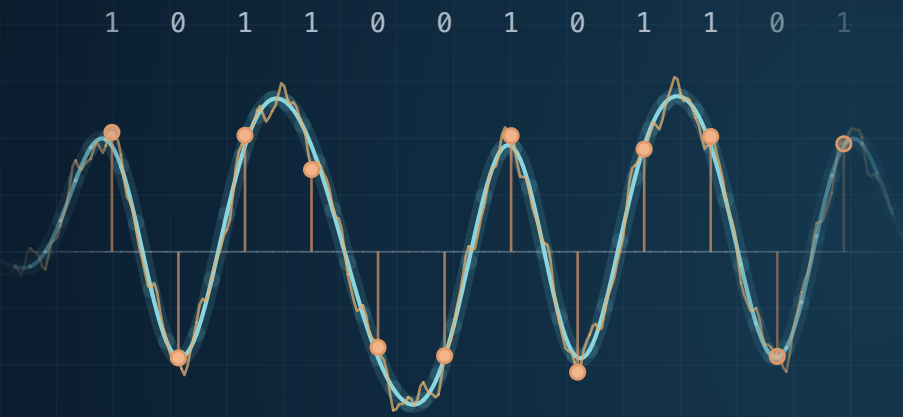




SAMPLING · DETECTION · MODULATION · CODING

# Digital Communications

*Formula and Notation Reference*



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CONVENTIONS · FORMULAS · NOTATION

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# Contents

*The conventions used throughout the course, every formula it establishes, and every symbol it defines. Nothing here is derived; the derivations are in the lecture notes.*

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## How to read it

Part 1 states the conventions, the two expressions every error probability comes from, and the transform pair Chapters 1 and 2 use. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

## PART

## 1

# Conventions

*These hold everywhere in the course, without local variation. Half the factor-of-two errors in this subject come from mixing two of them in one line, so where a result depends on one it is restated at that point rather than assumed.*

## CONVENTION STATEMENT

|                  |                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy and power | Normalised: the resistance is taken as $1 \Omega$ , so instantaneous power is $ x(t) ^2$ and energy is its integral. Signal energy is $E$ , energy per bit $E_b$ , energy per symbol $E_s$ , and $E_s = (\log_2 M) E_b$ . |
| Noise            | White and Gaussian with <b>two-sided</b> power spectral density $S_n(f) = N_0/2$ . Projected onto an orthonormal basis it gives independent components, each $\mathcal{N}(0, N_0/2)$ .                                    |
| The $Q$ function | $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ . It is the tail of a standard normal, so $Q(0) = \frac{1}{2}$ and $Q(-x) = 1 - Q(x)$ .                          |
| sinc             | Normalised: $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ , with zeros at every non-zero integer. Stated again wherever it is used.                                                                                      |
| Logarithms       | log without a base means base two, and every entropy and every bit count is in bits. Decibels are $10 \log_{10}$ of a power or energy ratio.                                                                              |
| Signals          | $s_i(t)$ is transmitted, $r(t)$ is received, $\{\psi_k(t)\}$ is the orthonormal basis, and boldface $\mathbf{s}_i$ is the point in signal space.                                                                          |
| Priors           | Symbols are equally likely unless a problem says otherwise, in which case the MAP form is used and the prior term written out.                                                                                            |

### THE TWO EXPRESSIONS THE WHOLE COURSE RESTS ON

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first is exact for two signal points and is where every error probability in Chapters 2, 4 and 5 comes from. The second is its nearest-neighbour extension to  $M$  points, and the 2 in the denominator is the noise variance per dimension being  $N_0/2$ .

## THE FOURIER TRANSFORM PAIR, AS CHAPTERS 1 AND 2 USE IT

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

Written in hertz rather than in rad/s, so no factor of  $2\pi$  appears in front of either integral. Sampling and the Nyquist criterion are both stated in this form.

## PART

## 2

# Summary of formulas

*The results each chapter carries forward, in course order and without derivations.*

## A.1 The transition from analog to digital — Chapter 1

The results of Chapter 1, with  $\text{sinc}(x) = \sin(\pi x)/(\pi x)$ .

| QUESTION                                            | RESULT                                                                                                                    | SOURCE                      |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| What does sampling do to the spectrum?              | $G_\delta(f) = f_s \sum_n G(f - nf_s)$ : a copy at every multiple of $f_s$ , scaled by $f_s$ .                            | Ch. 1 · PS<br>CH7.1.1       |
| What is the lowest safe sampling rate?              | The Nyquist rate $2W$ . Below it the copies overlap and alias.                                                            | Ch. 1 · PS<br>CH7.1.1       |
| What filter turns the samples back into the signal? | An ideal lowpass filter with gain $T_s$ . At $f_s = 2W$ its impulse response is $\text{sinc}(2Wt)$ .                      | Ch. 1 · PS<br>CH7.1.1       |
| How is $g(t)$ written in terms of its samples?      | $g(t) = \sum_n g(nT_s) \text{sinc}(2W(t - nT_s))$ , exact at every $t$ .                                                  | Ch. 1 · PS<br>CH7.1.1       |
| Mid-rise or mid-tread: which has a level at zero?   | Mid-tread. Mid-rise has a boundary at zero and outputs $\pm\Delta/2$ there.                                               | Ch. 1 · PS<br>CH7.2.1       |
| What is the power of the quantization noise?        | $E[Q^2] = \Delta^2/12$ , with $\Delta = 2m_{\max}/L$ , for a fine quantizer.                                              | Ch. 1 · PS<br>CH7.2.1       |
| What does one more bit buy?                         | 6.02 dB: $\text{SQNR} = \alpha + 6.02R$ , and $\alpha = 1.76$ dB for a full-scale sinusoid.                               | Ch. 1 · PS<br>CH7.2.1       |
| Why compand?                                        | Small amplitudes are common. Finer steps near zero keep the SQNR steady as the level falls.                               | Ch. 1 · PS<br>CH7.2.1       |
| What bit rate does PCM need?                        | $R_b = R f_s$ . Telephone speech: $8(8000) = 64$ kb/s.                                                                    | Ch. 1 · PS<br>CH7.3         |
| Why use a Gray code?                                | Adjacent levels differ in one bit, so a small decision error costs one bit error.                                         | Ch. 1 · PS<br>CH7.3         |
| What bandwidth does PCM need?                       | $B_T \geq R_b/2$ , which is $RW$ at $f_s = 2W$ . Each extra bit adds 6.02 dB and $W$ hertz.                               | Ch. 1 · PS<br>CH7.4.1       |
| What do DPCM and delta modulation send?             | The difference from a prediction. Delta modulation sends one bit a sample and needs $\Delta f_s$ above the largest slope. | Ch. 1 · PS<br>CH7.4.2–7.4.3 |

## A.2 Baseband transmission of digital signals — Chapter 2

The results of Chapter 2.

| QUESTION                                             | RESULT                                                                                                                                    | SOURCE                 |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Which filter gives the largest peak SNR?             | The matched filter $h(t) = k g(T - t)$ , or $H(f) = k G^*(f) e^{-j2\pi fT}$ , sampled at $t = T$ .                                        | Ch. 2 · PS<br>CH8.3.2  |
| What is that largest peak SNR?                       | $\eta_{\max} = 2E/N_0$ . It depends on the pulse energy $E$ , not on its shape.                                                           | Ch. 2 · PS<br>CH8.3.2  |
| Where do the two symbols of polar signalling sit?    | At $\pm\sqrt{E_b}$ on the axis of the unit-energy $\psi(t)$ , a distance $2\sqrt{E_b}$ apart.                                             | Ch. 2 · PS<br>CH8.2.1  |
| When do the correlator and the matched filter agree? | At $t = T_b$ , where both give $\int_0^{T_b} x(t) \psi(t) dt$ , for any shape of $\psi$ .                                                 | Ch. 2 · PS<br>CH8.3.1  |
| What noise is left in the demodulator output?        | $y = s_m + n$ , with $n$ Gaussian of mean 0 and variance $N_0/2$ .                                                                        | Ch. 2 · PS<br>CH8.3.1  |
| Where does the optimal threshold go?                 | $\lambda_{\text{opt}} = \frac{N_0}{4\sqrt{E_b}} \ln \frac{p_0}{p_1}$ , at 0 for equal priors and toward the less likely symbol otherwise. | Ch. 2 · PS<br>CH8.3.3  |
| What is $Q(x)$ ?                                     | The Gaussian tail $Q(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$ , the chance that a unit Gaussian exceeds $x$ .                            | Ch. 2 · PS<br>CH8.3.3  |
| What is $P_b$ for polar signalling?                  | $P_b = Q(\sqrt{2E_b/N_0})$ with equal priors and a matched filter.                                                                        | Ch. 2 · PS<br>CH8.3.3  |
| How much interference does an RC channel add?        | A bit $m$ bits back adds $(1 - q)q^m$ , with $q = e^{-2\pi BT_b}$ . The worst case totals $q$ .                                           | Ch. 2 · PS<br>CH10.1.1 |
| When does the eye close?                             | The opening is $2(1 - 2q)$ . It closes at $BT_b = \ln 2/(2\pi) = 0.110$ , and errors then occur without noise.                            | Ch. 2                  |
| What is Nyquist's criterion?                         | $p(kT_b) = \delta[k]$ exactly when $\sum_n P(f - nR_b) = T_b$ . It needs $B_T \geq R_b/2$ .                                               | Ch. 2 · PS<br>CH10.3.1 |
| What bandwidth does a raised cosine need?            | $B_T = \frac{R_b}{2}(1 + \alpha)$ . The tails fall as $1/ t ^3$ for $\alpha > 0$ .                                                        | Ch. 2 · PS<br>CH10.3.1 |

## A.3 Geometric representation of signal waveforms — Chapter 3

The results of Chapter 3.

| QUESTION                                         | RESULT                                                                                                                | SOURCE                |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|
| What is an orthonormal set?                      | Functions with $\int \psi_j \psi_k dt = 1$ for $j = k$ and 0 for $j \neq k$ : unit energy, and every pair orthogonal. | Ch. 3 · PS<br>CH8.1   |
| How is a coordinate computed?                    | $s_{ij} = \int_0^T s_i(t) \psi_j(t) dt$ , one correlator for each basis function.                                     | Ch. 3 · PS<br>CH8.1   |
| How is the waveform rebuilt?                     | $s_i(t) = \sum_{j=1}^N s_{ij} \psi_j(t)$ . The vector $\mathbf{s}_i$ holds the whole waveform.                        | Ch. 3 · PS<br>CH8.1   |
| What does the translation preserve?              | Inner products: $\int x(t) y(t) dt = \sum_k x_k y_k$ .                                                                | Ch. 3 · PS<br>CH8.1   |
| What is the energy of a signal?                  | $E_i = \ \mathbf{s}_i\ ^2$ , the squared distance from its point to the origin.                                       | Ch. 3 · PS<br>CH8.1   |
| What is the distance between two signals?        | $d_{ik} = \ \mathbf{s}_i - \mathbf{s}_k\ $ , the root of $\int (s_i - s_k)^2 dt$ .                                    | Ch. 3 · PS<br>CH8.1   |
| Antipodal or orthogonal: which is further apart? | Antipodal, $d = 2\sqrt{E_b}$ against $\sqrt{2E_b}$ . Orthogonal needs twice the energy, 3 dB, for the same distance.  | Ch. 3 · PS<br>CH8.2   |
| When are a cosine and a sine orthogonal?         | Over $[0, T]$ when $f_c T$ is a whole number. Scaled by $\sqrt{2/T}$ they are an orthonormal pair.                    | Ch. 3 · PS<br>CH8.6.1 |
| How far apart are the points of M-PSK?           | $d_{\min} = 2\sqrt{E} \sin(\pi/M)$ . All $M$ points lie on the circle of radius $\sqrt{E}$ .                          | Ch. 3 · PS<br>CH8.6.1 |
| What is one Gram–Schmidt step?                   | $g_k = s_k - \sum_{i < k} s_{ki} \psi_i$ , then $\psi_k = g_k / \sqrt{E_{g_k}}$ . A zero $g_k$ adds no axis.          | Ch. 3 · PS<br>CH8.1   |
| How many axes does a set of $M$ signals need?    | $N \leq M$ , with $N = M$ only when no signal is a combination of the others.                                         | Ch. 3 · PS<br>CH8.1   |
| What does another basis change?                  | Only the coordinates. $N$ , the energies and the distances stay, and so do the receiver and its error probability.    | Ch. 3 · PS<br>CH8.1   |

## A.4 The optimal receiver in additive white Gaussian noise — Chapter 4

The results of Chapter 4.

| RESULT       | STATEMENT                                                                                                                                                                                                         | SOURCE                |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| The receiver | $\hat{s} = \arg \max_i \left\{ \mathbf{r} \cdot \mathbf{s}_i - \frac{E_i}{2} + \frac{N_0}{2} \ln P(\mathbf{s}_i) \right\}$ . Correlate with each signal. Correct for energy and prior. Select the largest metric. | Ch. 4 · PS<br>CH8.4.1 |
| The picture  | $\hat{s} = \arg \min_i \ \mathbf{r} - \mathbf{s}_i\ ^2$ . Nearest-point boundaries are perpendicular bisectors. Unequal priors change the region sizes.                                                           | Ch. 4 · PS<br>CH8.4.1 |
| Binary       | $P_e = Q\left(\sqrt{d^2/2N_0}\right)$ . For two points, the distance gives the exact error probability.                                                                                                           | Ch. 4 · PS<br>CH8.3.3 |
| M-ary        | $P_e \approx N_{\min} Q\left(\sqrt{d_{\min}^2/2N_0}\right)$ . For larger constellations, nearest neighbours give the dominant error terms.                                                                        | Ch. 4 · PS<br>CH8.4.2 |

## A.5 Digital modulation methods — Chapter 5

The results of Chapter 5.

| QUESTION                                    | RESULT                                                                                                   | SOURCE                         |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------|
| What does a carrier do to the spectrum?     | $U(f) = \frac{1}{2}S(f - f_c) + \frac{1}{2}S(f + f_c)$ . The band doubles to $2W$ and the energy halves. | Ch. 5 · PS CH8.5.1             |
| What does the IQ modulator send?            | $s(t) = I \cos(2\pi f_c t) - Q \sin(2\pi f_c t)$ : one point of the plane a symbol.                      | Ch. 5 · PS CH8.6, 8.7          |
| How do the binary schemes compare?          | BPSK: $Q(\sqrt{2E_b/N_0})$ . BFSK and BASK: $Q(\sqrt{E_b/N_0})$ , 3 dB worse.                            | Ch. 5 · PS CH8.3.3, 8.6.1, 9.5 |
| What is the M-PSK neighbour distance?       | $d_{\min} = 2\sqrt{E_s} \sin(\pi/M)$ , and $P_e \approx 2Q(\sqrt{2E_s/N_0} \sin(\pi/M))$ .               | Ch. 5 · PS CH8.6.1, 8.6.3      |
| What do Gray labels buy?                    | Neighbours differ in one bit, so $P_b \approx P_e / \log_2 M$ .                                          | Ch. 5 · PS CH8.6.1, 8.6.3      |
| What does DPSK save, and what does it cost? | No carrier phase. $P_b = \frac{1}{2}e^{-E_b/N_0}$ , under 1 dB behind BPSK.                              | Ch. 5 · PS CH8.6.4, 8.6.5      |
| What is the M-ASK error?                    | $\frac{2(M-1)}{M}Q(\sqrt{6E_s/((M^2-1)N_0)})$ , about 6 dB a doubling.                                   | Ch. 5 · PS CH8.5.3             |
| What is the square-QAM error?               | $4(1 - 1/\sqrt{M})Q(\sqrt{3E_s/((M-1)N_0)})$ , with $E_s = (M-1)d^2/6$ .                                 | Ch. 5 · PS CH8.7.3             |
| How much does QAM save over PSK?            | 4.20 dB at $M = 16$ and 9.95 dB at $M = 64$ .                                                            | Ch. 5 · PS CH8.7.3             |
| What do M orthogonal signals give?          | $M - 1$ neighbours at $\sqrt{2E_s}$ . The $E_b/N_0$ needed falls as $M$ grows, toward $-1.6$ dB.         | Ch. 5 · PS CH9.1.1, 9.1.2      |
| What does noncoherent BFSK need?            | Tones $1/T$ apart and an envelope detector. $P_b = \frac{1}{2}e^{-E_b/2N_0}$ .                           | Ch. 5 · PS CH9.5.2, 9.5.3      |
| What is MSK?                                | BFSK at $\Delta f = 1/(2T_b)$ with a continuous phase. Its envelope is constant.                         | Ch. 5 · PS CH9.6.1, 9.6.2      |
| How wide is each family?                    | PSK and QAM: $R_b / \log_2 M$ . Orthogonal: $MR_b / (2 \log_2 M)$ .                                      | Ch. 5 · PS CH10.2, 9.7         |
| Where does each family sit on the plane?    | PAM, PSK and QAM: bandwidth-limited, $r > 1$ . Orthogonal: power-limited, $r < 1$ .                      | Ch. 5 · PS CH9.7               |

## A.6 An introduction to information theory — Chapter 6

The results of Chapter 6.

| QUESTION                                            | RESULT                                                                                               | SOURCE                 |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------|
| How much information does a symbol carry?           | $I(s_k) = -\log_2 p_k$ bits: rare symbols carry more.                                                | Ch. 6 · PS<br>CH12.1.1 |
| What is entropy, and what bounds it?                | $H(S) = -\sum_k p_k \log_2 p_k$ , with $0 \leq H \leq \log_2 K$ .                                    | Ch. 6 · PS<br>CH12.1.1 |
| What does an extended source carry?                 | $H(S^n) = nH(S)$ for a memoryless source.                                                            | Ch. 6 · PS<br>CH12.1.2 |
| What are typical sequences?                         | About $2^{nH}$ sequences, each of probability near $2^{-nH}$ , that hold almost all the probability. | Ch. 6 · PS<br>CH12.2   |
| When do codeword lengths allow a prefix code?       | Exactly when $\sum_k 2^{-l_k} \leq 1$ .                                                              | Ch. 6                  |
| How close can a code get to the entropy?            | $H \leq \bar{L} < H + 1$ , and $H + 1/n$ for blocks of $n$ .                                         | Ch. 6 · PS<br>CH12.3.1 |
| How is a Huffman code built?                        | Merge the two least likely entries until one is left, then read each codeword back.                  | Ch. 6 · PS<br>CH12.3.1 |
| What does Lempel–Ziv need to know about the source? | Nothing. It parses the stream into new phrases, and its cost falls toward $H$ as the stream grows.   | Ch. 6 · PS<br>CH12.3.2 |
| What is a binary symmetric channel?                 | Each bit flips with probability $p$ . Hard-decision BPSK gives $p = Q(\sqrt{2E_b/N_0})$ .            | Ch. 6 · PS<br>CH12.4   |
| What is mutual information?                         | $I(X; Y) = H(X) - H(X   Y)$ , symmetric and never negative.                                          | Ch. 6 · PS<br>CH12.1.3 |
| What is the capacity of a channel?                  | $C = \max_{p(x)} I(X; Y)$ : $1 - H_b(p)$ for the BSC, $1 - \epsilon$ for the erasure channel.        | Ch. 6 · PS<br>CH12.5   |
| What does the channel coding theorem say?           | Rates below $C$ can be made reliable. Rates above it cannot.                                         | Ch. 6 · PS<br>CH12.5   |
| What is the capacity of a bandlimited channel?      | $C = W \log_2(1 + P/N_0 W)$ b/s, which levels off at $1.44 P/N_0$ as $W$ grows.                      | Ch. 6 · PS<br>CH12.5.1 |
| What is the Shannon limit?                          | $E_b/N_0 > \ln 2 = -1.59$ dB. Uncoded BPSK sits 11.2 dB above it.                                    | Ch. 6 · PS<br>CH12.6   |

## PART

## 3

# Notation

---

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

$f_s, T_s$

Sampling rate in hertz and the interval between samples,  $T_s = 1/f_s$ .

$W$

The highest frequency present in a message. The sampling theorem asks for  $f_s > 2W$ .

$\text{sinc}(x)$

Normalised:  $\sin(\pi x)/(\pi x)$ , with zeros at every non-zero integer.

$\Delta, L, n$

Quantizer step, number of levels and bits a sample, with  $L = 2^n$  and  $\Delta = 2V/L$  over a range  $\pm V$ .

$\sigma_q^2$

Quantization noise power,  $\Delta^2/12$  for a uniform quantizer.

$T, R$

Symbol interval and the rate that follows from it. A bit interval is written  $T_b$  and a bit rate  $R_b$ .

$p(t), h(t)$

The transmitted pulse and the impulse response of the receive filter. The matched filter is  $h(t) = p(T - t)$ .

$\alpha$

Roll-off of a raised-cosine pulse, from 0 to 1. Bandwidth is  $(1 + \alpha)/2T$ .

$\psi_k(t)$

The  $k$ -th orthonormal basis function, from the Gram–Schmidt procedure applied to the signal set.

$s_i(t), \mathbf{s}_i$

The  $i$ -th transmitted waveform, and the point in signal space whose coordinates are its projections onto the basis.

$E, E_i$

Signal energy, normalised so that no resistance appears:  $E_i = \|\mathbf{s}_i\|^2$ .

$d_{ij}, d_{\min}$

Distance between two signal points, and the smallest such distance in a constellation.

$r(t), \mathbf{r}$

The received waveform and the observation vector it produces,  $\mathbf{r} = \mathbf{s}_i + \mathbf{n}$ .

$N_0$

Noise power spectral density parameter. The two-sided density is  $N_0/2$  watts per hertz, and each noise component is  $\mathcal{N}(0, N_0/2)$ .

$Q(x)$ 

The Gaussian tail: the probability that a standard normal variable exceeds  $x$ . Equal to  $\frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ .

 $M, N$ 

The number of signals in the set, and the number of basis functions they need.  $M$  signals carry  $\log_2 M$  bits a symbol.

 $N_{\min}$ 

The average number of signal points at the minimum distance. An average, so not usually a whole number.

 $P_b, P_e, P_s$ 

Bit error probability, error probability in general, and symbol error probability.

 $E_b, E_s$ 

Energy per information bit and energy per transmitted symbol, with  $E_s = (\log_2 M) E_b$ .

 $f_c$ 

Carrier frequency. Every scheme in Chapter 5 changes the amplitude, the phase or the frequency of  $\cos(2\pi f_c t)$ .

 $S, K, p_k$ 

The source alphabet, how many symbols it has, and the probability of the  $k$ -th.

 $I(s_k)$ 

Self-information of one symbol,  $-\log_2 p_k$  bits.

 $H(S)$ 

Entropy: the average information a symbol carries, in bits a symbol. Bounded by 0 and  $\log_2 K$ .

 $\bar{L}, l_k$ 

Average codeword length and the length of the codeword for symbol  $k$ .

 $\eta$ 

Coding efficiency,  $H(S)/\bar{L}$ . Never above one.

 $\sigma^2$ 

Variance of the codeword length about  $\bar{L}$ . Separates two Huffman codes that share an average.

 $p(y_k | x_j)$ 

Transition probability of a discrete memoryless channel. Collected into the channel matrix, whose every row sums to one.

 $H(X | Y)$ 

Conditional entropy: the uncertainty still remaining about the input after the output has been seen.

 $I(X; Y)$ 

Mutual information,  $H(X) - H(X | Y)$ : what the channel actually delivered, in bits per use. Symmetric and never negative.

 $C$ 

Channel capacity, the largest  $I(X; Y)$  over all input distributions. For the binary symmetric channel,  $1 - H(p)$ .

 $H(p)$ 

The binary entropy function,  $-p \log_2 p - (1 - p) \log_2 (1 - p)$ . It measures a two-symbol source and also the noise a binary symmetric channel adds.

**Digital Communications · Formula and Notation Reference**

| VERSION | DATE              | CHANGES        |
|---------|-------------------|----------------|
| v0      | 24 September 2026 | First edition. |

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